MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 12: Complex Series

Nick Strigul
Washington State University Vancouver
nick.strigul@wsu.edu

February 17, 2023

Lecture outline

1. Complex Series
2. Convergence
3. Basic theorems
4. Power Series

Complex series

Consider a series of complex numbers $\alpha_1 + \alpha_2 + \alpha_3 + \cdots$. Let

$$A_n = \alpha_1 + \alpha_2 + \alpha_3 + \cdots + \alpha_n$$

A_n is called the *nth* partial sum.

Definition

The series $\sum_{n=1}^{\infty} \alpha_n$ is said to *converge* to the number A if the sequence of partial sums $\{A_n\}$ converges to A.

Remark: Thus

$$\sum_{n=1}^{\infty} \alpha_n$$

is said to *converge* to the number A if given $\varepsilon > 0$, $\exists$ a positive integer N (which depends on ε) such that $|A - A_n| < \varepsilon$ whenever $n > N(\varepsilon)$. We write

$$\sum_{n=1}^{\infty} \alpha_n = A.$$

Convergence of complex series

Theorem

Let $\alpha_n = b_n + ic_n$, where b_n and c_n are real. Then $\sum_{n=1}^{\infty} \alpha_n$ converges $\Longleftrightarrow$ $\sum_{n=1}^{\infty} b_n$ and $\sum_{n=1}^{\infty} c_n$ converge.

Proof: Suppose $\sum_{n=1}^{\infty} \alpha_n = A$ where A is complex. Write $A = B + iC$, and $A_n = B_n + iC_n$. By definition given $\varepsilon > 0$ $\exists$ an integer $N(\varepsilon)$ such that

$$|A - A_n| = |(B + iC) - (B_n + iC_n)| = |(B - B_n) + i(C - C_n)| < \varepsilon$$

Then certainly $|B - B_n| < \varepsilon$ and $|C - C_n| < \varepsilon$ for $n > N$. Hence $\sum b_n$ and $\sum c_n$ converge.

Now suppose that $\sum b_n = B$ and $\sum c_n = C$. By definition given $\varepsilon > 0$ $\exists$ $N(\varepsilon)$ such that

$$|B - B_n| < \varepsilon/2 \text{ and } |C - C_n| < \varepsilon/2 \quad \text{for } n > N.$$

Thus $|(B + iC) - (B_n + iC_n)| \leq |B - B_n| + |C - C_n| < \varepsilon$. Therefore the sequence $\{B_n + iC_n\} = A_n$ converges to $\{B + iC\} = A$.

Cauchy series

Definition

(Cauchy or Intrinsic Convergence) The series $\sum_{n=1}^{\infty} \alpha_n$ is said to converge *intrinsically* (or in the Cauchy sense) if given arbitrary $\varepsilon > 0$ $\exists$ a positive integer $N = N(\varepsilon)$ such that $|A_n - A_n| < \varepsilon$ whenever $n > N$ and $m > N$.

Theorem

$\sum_{n=1}^{\infty} \alpha_n$ *is convergent* $\iff$ *it is Cauchy convergent.*

Remark: If $\sum \alpha_n$ converges, then $|\alpha_n| \longrightarrow 0$ as $n \to \infty$. The converse is not true. Consider $\sum_1^{\infty} \frac{1}{n}$.

Proof of Cauchy convergent $\Longrightarrow$ convergent

Suppose that $\sum \alpha_n$ is Cauchy convergent. That is given $\varepsilon > 0$ $\exists$ $N(\varepsilon)$ such that $|A_n - A_{n+p}| < \varepsilon$ when $n > N$ for $p = 1, 2, \ldots$ Therefore the numbers A_{n+p} all lie within an ε neighborhood of A_n. By the Bolzano-Weierstrass Theorem (Each infinite bounded sequence in $\mathbb{R}^n$ has a convergent subsequence) $\exists$ at least one limit point. Suppose there are two limit points A and A' . Let $|A - A'| = d > 0$. Within any $d/3$ neighborhood of A and A' $\exists$ an infinite subsequence of points such that $A_{n_i} \to A$ and $A_{m_i} \to A'$ where n_i and m_i
go to ∞ as $i \longrightarrow \infty$.
Then the Cauchy condition is contradicted. For set $\varepsilon = d/3$. Then $\exists$ infinitely many pairs of points (A_{n_i}, A_{m_i}) such that

$$|A_{m_i} - A_{n_i}| \geq d/3 = \varepsilon$$

Hence $\exists$ exactly one limit point of the sequence $\{A_n\}$.

Absolute convergence

Definition

$\sum \alpha_n$ is said to be *absolutely convergent* if the real series $\sum |\alpha_n|$ converges.

Remark: absolute convergence $\Longrightarrow$ (ordinary) convergence since

$$|\alpha_{m+1} + \cdots\cdots + \alpha_n| \leq |\alpha_{m+1}| + \cdots\cdots |\alpha_n|$$

so that $|A_n - A_m| \leq |D_n - D_m|$ where $|D_m| = |\alpha_1| + \cdots + |\alpha_m|$

Uniform convergence 1

Suppose we have functions $f_1(z), f_2(z), \ldots$ defined in R. Then for each $z \in R$

$$\sum_{n=1}^{\infty} f_n(z) \qquad (1)$$

is a series of constants. Suppose that (1) converges for all $z \in R$. Consider z_1. Given $\varepsilon > 0$ $\exists$ $N(\varepsilon, z_1)$ such that .... If we can find an N that works for all z, then we have uniform convergence.

Let

$$S_n(z) = \sum_{i=1}^{n} f_i(z)$$

Definition

The series $\sum_{n=1}^{\infty} f_n(z)$ is said to *converge uniformly* to the function $f(z)$ if, given an arbitrary $\varepsilon > 0$, $\exists$ a positive integer $N = N(\varepsilon)$ such that

$$|f(z) - S_n(z)| < \varepsilon$$

for all $n > N$ and all $z \in R$.

Uniform convergence 2

Remark: If the convergence is *not* uniform, then $N = N(\varepsilon, z)$. That is, given $\varepsilon > 0$ then for any finite N you can find a $z \varepsilon R$ such that

$$|f(z) - S_n(z)| \geq \varepsilon$$

for some $n > N$.

Example. Part 1

Example

The series $\sum_{n=1}^{\infty} z^{n-1}$ converges for all z such that $|z| < 1$, but the convergence is not uniform on the domain.

Convergence at a point z : Pick z such that $|z| = r < 1$. Now
$S_n = 1 + z + z^2 + \cdots + z^{n-1} = \dfrac{1 - z^n}{1 - z} = \dfrac{1}{1 - z} - \dfrac{z^n}{1 - z}$.
Thus

$$\left| \frac{1}{1 - z} - S_n(z) \right| = \frac{|z|^n}{|1 - z|} = \frac{r^n}{1 - r} < \varepsilon$$

for $n >$ some suitable N, since $r < 1$.

Example. Part 2

Example

Nonuniform convergence: Given $\varepsilon > 0$ assume $\exists$ an $N = N(\varepsilon)$ etc.

$$|f(z) - S_n(z)| = \frac{|z|^n}{|1-z|}$$

Choose $n > N$ (say, $n = N+1$). Suppose $|z| > 1/2$, so that

$$\frac{|z|^N}{|1-z|} > \frac{\left(\frac{1}{2}\right)^{N+1}}{|1-z|}$$

Take $z = x$, a real number, where $\frac{1}{2} < x < 1$. Then

$$\frac{\left(\frac{1}{2}\right)^{N+1}}{1-x} > \varepsilon$$

if x is sufficiently close to 1.

Series of continuous functions

Theorem

If $f_1(z)$, $f_2(z)$, ...$f_n(z)$, are all continuous in R and if $\sum_{i=1}^{\infty} f_i(z)$ converges uniformly in R to $f(z)$, then the sum $f(z)$ is again a continuous function in R.

Proof: We write $f(z) = \sum_{i=1}^{n} f_i(z) + \sum_{i=n+1}^{\infty} f_i(z) = S_n(z) + r_n(z)$.
Suppose $z_0, z_1 \in R$. Then

$$f(z_0) - f(z_1) = ([S_n(z_0) - S_n(z_1)] + [r_n(z_0) - r_n(z_1)])$$

$$\Longrightarrow \qquad |f(z_0) - f(z_1)| \leq |S_n(z_0) - S_n(z_1)| + |r_n(z_0) - r_n(z_1)|$$

Given $\varepsilon > 0$ $\exists$ $N = N(\varepsilon)$ such that for any $z \in R$ $\quad |r_n(z)| < \varepsilon$ for $n > N$. Hence

$$|f(z_0) - f(z_1)| < |S_n(z_0) - S_n(z_1)| + 2\varepsilon \quad \text{for } n > N \qquad (1)$$

Now fix n. (1) holds for all z_0, z_1 in R. Now $S_n(z)$ is a continuous function so $\exists$ a $\delta > 0$ such that $|S_n(z_0) - S_n(z_1)| < \varepsilon$ if $|z_0 - z_1| < \delta$.
Now fix z_0, prescribe ε, choose N, and then pick δ. Then

$$|f(z_0) - f(z_1)| < 3\varepsilon$$

whenever $|z_0 - z_1| < \delta$. Hence $f(z)$ is continuous at z_0.

Power Series. Part 1.

Consider

$$\sum_{n=0}^{\infty} a_n z^n \qquad (1)$$

where $a_0, a_1, \ldots$ are complex constants. Similar results will hold for

$$\sum_{n=0}^{\infty} a_n (z-a)^n \qquad (2)$$

when we substitute $z - a$ for z.

Remark: (1) converges at least for $z = 0$, whereas (2) converges at least for $z = a$. Thus there is always at least one point at which a power series converges.

Remark: Recall that $\sum_{n=0}^{\infty} r^n$ converges for $|r| < 1$, r real. This series is called the geometric series and

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

Power Series Convergence Theorem

Theorem

Suppose that $\sum_{n=0}^{\infty} a_n(z-a)^n$ *converges for some* z_0 *different from* a*. Then this series converges absolutely* $\forall\ z$ *in the open disk* $|z-a| < |z-z_0|$*. In any concentric closed disk,* $|z-a| \leq \rho < |z-z_0|$ *the convergence is uniform.*

Remark: Weierstrass M Test: If in a region R $|u_n(z)| \leq M_n$, $n = 1, 2, 3, \ldots$ where M_n are positive constants such that $\sum M_n$ converges, the $\sum u_n(z)$ is uniformly convergent.

Proof of Power Series Convergence Theorem

Proof: Assume $\sum_{n=0}^{\infty} a_n(z_0 - a)^n$ converges. Then $\lim_{n\to\infty} a_n(z_0 - a)^n = 0$. Thus for some N,

$$|a_n(z_0 - a)^n| < 1$$

if $n \geq N$. For $n \geq N$ we have

$$|a_n(z - a)^n| = |a_n| \left|\frac{z - a}{z_0 - a}\right|^n |z_0 - a|^n \leq \left|\frac{z - a}{z_0 - a}\right|^n \qquad (*)$$

But $\sum_0^{\infty} \left|\frac{z - a}{z_0 - a}\right|^n$ is a geometric series with $r = \left|\frac{z - a}{z_0 - a}\right| < 1$. Therefore by comparison

$$\sum_{n=N}^{\infty} |a_n(z - a)^n|$$

converges. Hence $\sum_{n=0}^{\infty} |a_n(z - a)^n|$ also converges.
The proof that the convergence is uniform follows from $(*)$ for $|z - a| \leq \rho < |z - z_0|$ and the Weierstrass M Test.

Corollary of Power Series Convergence Theorem

Corollary

Either (a)

$$\sum_{n=0}^{\infty} a_n (z-a)^n$$

converges for all finite z or (b) $\exists$ a real number $R > 0$ such that

$$\sum_{n=0}^{\infty} a_n (z-a)^n$$

converges for all z with $|z-a| < R$ and diverges for $|z-a| > R$.

Note: R is called the radius of convergence of the power series.

Proof: Without loss of generality we assume that $a = 0$. If (a) is not true, then $\exists$ a value of z, say z_1, where (1) diverges. Let S be the set of all real numbers $|z|$ such that $\sum a_n z^n$ converges at z. From the above theorem, $|z_1|$ is an upper bound for S. Since S is a bounded set, it has a least upper bound, which will be R.

Example of Power Series. Part 1

Example

Consider $\sum_{n=0}^{\infty} \left(\frac{2}{3i}\right)^n (z-i)^n$. We want to know the radius of convergence of this series.

Recall the Ratio test for real series, namely suppose that if $b_n \neq 0$ for $n = 0, 1, 2, \ldots$ and that

$$\lim_{n\to\infty} \left|\frac{b_{n+1}}{b_n}\right| = L$$

in which L may equal ∞. Then the series $\sum_{n=0}^{\infty} b_n$ converges absolutely if $L < 1$ and diverges if $L > 1$. The test gives no conclusion if $L = 1$.

Example of Power Series. Part 2

Example

Taking the absolute value of the ratio of successive terms yields;

$$\left|\frac{(2/3i)^{n+1}(z-i)^{n+1}}{(2/3i)^{n}(z-i)^{n}}\right| = \tfrac{2}{3}|z-i| \to \tfrac{2}{3}|z-i| \text{ as } n \to \infty$$

Thus the series converges absolutely if this limit is less than 1, that is if $|z-i| < 3/2$ and diverges if the limit is greater than 1, that is if $|z-i| > 3/2$. The radius of convergence for this power series is $3/2$ and the open disk of convergence is the open disk of radius $3/2$ centered at i.

Example: boundary behaviour

Example

Boundary Behavior: In the following $R = 1$.
1) $\sum_0^\infty z^n$ converges for *no* boundary points.
2) $\sum_1^\infty \frac{z^n}{n^2}$ converges for all boundary points since $\left|\frac{z^n}{n^2}\right| \leq \frac{1}{n^2}$ because $|z| \leq 1$.
3) $\sum_1^\infty \frac{z^n}{n}$ converges for some, but not all, of its boundary points. It diverges for $z = 1$ and converges for $z = -1$.

Lemma: $\sum_{n=0}^{\infty} a_n(z-a)^n$ and $\sum_{n=1}^{\infty} na_n(z-a)^{n-1}$. Part 1

Lemma

The power series $\sum_{n=0}^{\infty} a_n(z-a)^n$ and $\sum_{n=1}^{\infty} na_n(z-a)^{n-1}$ have the same radius of convergence.

Note: Both series are power series, and the second is obtained from the first by differentiating term by term.

Proof: Let the first series have radius of convergence R and the second R^*.

First suppose that $|z_0 - a| < R^*$ so that $\sum_{n=1}^{\infty} |na_n(z-a)^{n-1}|$ converges. Now

$$\left|a_n (z_0 - a)^n\right| \leq n\,|z_0 - a| \left|a_n (z_0 - a)^{n-1}\right|$$

for $n = 1, 2, \ldots$ But $\sum_1^{\infty} n\,|z_0 - a|$
$\left|a_n (z_0 - a)^{n-1}\right| = |z_0 - a| \sum_1^{\infty} \left|na_n (z_0 - a)^{n-1}\right|$
that is, a constant times a convergent series. By comparison $\sum_0^{\infty} \left|a_n (z_0 - a)^n\right|$ converges. Therefore wherever the second series converges, so does the first. We conclude that $R^* \leq R$.

Lemma: $\sum_{n=0}^{\infty} a_n(z-a)^n$ and $\sum_{n=1}^{\infty} na_n(z-a)^{n-1}$. Part 2

We now show that $R \leq R^*$. Suppose that $|z_0 - a| < R$. We will prove that the second series converges absolutely at z_0. We may assume that $z_0 \neq a$. Choose r with $|z_0 - a| < r < R$. Then

$$\left| na_n (z_0 - a)^{n-1} \right| = \frac{n}{|z_0 - a|} \left(\frac{|z_0 - a|}{r} \right)^n |a_n| \, r^n \qquad (1)$$

Now

$$\sum_1^{\infty} n \left(\frac{|z_0 - a|}{r} \right)^n \qquad (2)$$

converges since using the ratio test we have

$$\frac{(n+1)\left(\frac{|z_0 - a|}{r} \right)^{n+1}}{n\left(\frac{|z_0 - a|}{r} \right)^n} = \frac{n+1}{n} \frac{|z_0 - a|}{r} \to \frac{|z_0 - a|}{r}$$

as $n \to \infty$, and this quotient is less than 1 by our choice of r. Since (2) converges there exists a number M such that $n \left(\frac{|z_0 - a|}{r} \right)^n \leq M$ for $n = 1, 2, \ldots$

Lemma: $\sum_{n=0}^{\infty} a_n(z-a)^n$ and $\sum_{n=1}^{\infty} na_n(z-a)^{n-1}$. Part 3

Thus (1) $\Longrightarrow$

$$\left|na_n\left(z_0-a\right)^{n-1}\right| \leq \frac{M}{|z_0-a|}\left|a_n\right| r^n$$

for $n=1,2,\ldots$ But $\sum_0^{\infty}\left|a_n\right| r^n$ convergres since $r<R$. Therefore by the comparison test $\sum_1^{\infty}\left|na_n\left(z_0-a\right)^{n-1}\right|$ also converges. Therefore $R \leq R^*$, completing the proof.

Note that the argument remains valid if the series have infinite radii of convergence.

Radius of convergence theorem.

Theorem

Let $\sum_0^\infty a_n(z-a)^n$ *have radius of convergence* R*, where* $0 < R \leq \infty$*. Let*

$$f(z) = \sum_{n=0}^\infty a_n(z-a)^n$$

for $|z-a| < R$*. Then* f *is differentiable for* $|z-a| < R$ *and*

$$f'(z) = \sum_{n=1}^\infty na_n(z-a)^{n-1}$$

Proof: The proof involves letting $g(z) = \sum_{n=1}^\infty na_n(z-a)^{n-1}$ and looking at

$$\left|\frac{f(z+h)-f(z)}{h} - g(z)\right| = \left|\frac{1}{h}\sum_0^\infty a_n\left((z+h)^n - z^n\right) - \sum_{n=1}^\infty na_n(z-a)^{n-1}\right|$$

as $h \to 0$. One must then use the binomial theorem and other estimates. We omit the details.

Power Series: Summary

We may summarize the results we have as follows:

Theorem

In the interior of its circle of convergence
a) a power series represents an analytic function of z,
and
b) its derivative can be found by term by term differentiation.

Corollary

A power series has derivatives of all orders. These are again power series with the same circle of convergence as the original series.

Taylor series and Power series function

Corollary

A power series is the Taylor series of the function represented by it.

Proof: For if

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n$$

for $|z-a| < R$, then differentiating n times and setting $z = a$ gives

$$f^{(n)}(a) = a_n n!$$

so that

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z-a)^n$$

Thank you